v1.0.15 — Multi-line + paste-IN + Claude Code TUI + workable-items SQLite + DOCX + constitution sync

Revision: 1 **Last modified:** 2026-05-28T15:00:00Z **Released:** 2026-05-28 **versionCode:** 16 **Authority:** constitution submodule `Constitution.md` §11.4 + project `Constitution.md` §101

Triggering operator mandate (verbatim, 2026-05-28)

> "In last iteration of work we did we have allegedly fixed copy / paste from and to the tmux session window. Selecting multiple lines and copying of them does not work. We MUST BE able to scroll vertically everywhere and copy / paste anything! Especially in Claude Code (claude command)! Do in depth investigation and systematic debugging, reproduce issue and then apply comprehensive fixes! All fixes MUST BE ALWAYS covered with all supported types of the tests we have in the project! Full automation and e2e tests particularly MUST ALWAYS produce real (physical) proof(s) of changes and fixes we did working as expected! Having false positives (or negatives) or any bluff(s) IS STRICTLY FORBIDDEN! ... We MUST for every issue or change, for any workable item create proper workable item in our Issues doc and all relevant docs around it (with them all being exported in all expected file types - PDF, HTML, DOCX)! We MUST NEVER forget the flow: workable item → SQLite database → all docs we have related to workable items!"

Follow-up (verbatim, 2026-05-28):

> "Once all done, make sure we execute setup script and current host machine is fully updated, and validated and tested. Then do all this for nezha host like we used to do already many times!"

Follow-up (verbatim, 2026-05-28):

> "When everything is fully tested and validated create new release version using GitHub and GitLab CLIs. Do not forget to create comprehensive version (change) log for the release! After releasing and pushing everything to all upstreams, increase version (version code) and commit and push all work - all submodules and main repo to all upstreams!"

Scope (5-PWU pipeline)

This release executed five Parallel Work Units per §11.4.58 + §11.4.94 zero-idle parallel-by-default:

PWU	Subject	Status
Phase 0	`constitution/` ff-merge 84c948d → 6828ff2 + `Containers/` fbef9d6 → 2e9ca0e (§11.4.37 / §11.4.26)	✓
A	tmux copy/paste fix + 4 tests (45/46/47/48) + helper + 4-layer coverage	✓
B	§11.4.87–98 short-form propagation to project governance	✓
C	SQLite workable-items Go binary (FULL Phase 3+ project-local)	✓
D	DOCX export extension to markdown sync pipeline	✓
E	Issues→DB→docs round-trip, commit/push, multi-host deploy, release	IN PROGRESS

Highlights — the user's primary copy/paste ask

Multi-line drag selection inside Claude Code

Claude Code (and any modern TUI that requests `CSI ?1003h` mouse tracking) sets `#{mouse_any_flag}=1` on the active pane. tmux's DEFAULT `MouseDrag1Pane` binding then forwards the drag to the app via `send -M` — the operator's drag never reaches tmux copy-mode, so no selection happens.

v1.0.15 adds a dedicated modifier path:

```
bind -n M-MouseDrag1Pane copy-mode -M
bind -n S-MouseDrag1Pane copy-mode -M

bind -T copy-mode-vi M-MouseDragEnd1Pane send-keys -X copy-pipe-and-cancel "#{@clip}"
bind -T copy-mode-vi S-MouseDragEnd1Pane send-keys -X copy-pipe-and-cancel "#{@clip}"
```

Hold **Alt** (macOS Terminal.app / iTerm2 / WezTerm) or **Shift** (Linux convention) while dragging to force tmux to take the drag. A plain unmodified drag still goes to the app — preserving native app interactivity.

Paste-INTO tmux from the OS clipboard

tmux's `set-clipboard external` is COPY-OUT only (OSC-52 paste is not a real thing). v1.0.15 adds the missing READ counterpart:

```
set -g @clip-read 'sh -c "command -v pbpaste >/dev/null 2>&1 && exec pbpaste; command -v wl-paste >/dev/null 2>&1 && exec wl-paste
```

```
-n; command -v xclip >/dev/null 2>&1 && exec xclip -o -selection
clipboard; command -v termux-clipboard-get >/dev/null 2>&1 && exec
termux-clipboard-get; printf \"\""'

bind P run -b 'tmux load-buffer - <<< "$(#{@clip-read})" \; tmux
paste-buffer -p'
```

Press `prefix + P` (capital — lowercase `p` is previous-window) to paste the OS clipboard contents into the active pane.

Multi-line keyboard copy

Already worked on v1.0.14 (using `v` + `cursor-down -N <count>` + `end-of-line` + `y`) but had no test coverage. Test 45 is the regression-guard: `printf` 6 markers, select 6 lines, copy via `@clip` pipe, read `pbpaste` back — physical multi-line proof.

Scroll vertically inside Claude Code

Already worked on v1.0.3 via the `bind -n WheelUpPane` override. Test 47 finally asserts this end-to-end inside an alt-screen + mouse-tracking surface using `helpers/synthetic_alt_screen_app.py` (no OAuth flake per §11.4.98).

Highlights — governance + plumbing

Constitution submodule sync (PWU-B)

Pulled 19 new commits from `HelixDevelopment/HelixConstitution` `main` (`6828ff2`). New universal anchors §11.4.87–§11.4.98 short- form-propagated into project Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md, with each anchor token appearing ≥3 times in each consumer per PWU-B audit. New verify.sh Layer-1 gates `CM-COVENANT-114-87..98-PROPAGATION` (12 gates) enforce literal presence.

`Containers/` submodule advanced 17 commits (`fbef9d6` → `2e9ca0e`).

Workable-items SQLite SSoT (PWU-C)

The constitution at `6828ff2` introduces §11.4.93/95 — a SQLite- backed single-source-of-truth for workable items at `docs/workable_items.db`, driven by a Go binary with sync/diff/ validate/add/close/report subcommands. The constitution scaffold ships **Phase-2 stubs only** (every subcommand exits code 2). Per operator decision `feedback_no_modify_constitution`, the project cannot modify the constitution to implement Phase 3+. PWU-C implements Phase 3+ **project-local** at `cmd/workable-items/`:

- 11 Go sources + embedded `schema.sql` (verbatim copy from constitution `6828ff2` with `-- DRIFT-CHECK:` header)
- Pure-Go `modernc.org/sqlite` — no CGO so cross-compile to Mistborn arm64 + nezha x86_64 works without toolchain dance

- 10 unit + round-trip tests pass 30/30 (3 iters per §11.4.50 deterministic-consistency)
- Initial DB seeded from live `Issues.md` (1 item) + `Fixed.md` (44 items) → 45 ATM-NNN allocations (ATM-001..ATM-045)
- DB TRACKED in git per §11.4.95 explicit carve-out from §11.4.30

DOCX export (PWU-D)

`scripts/sync_all_markdown_exports.sh` emits `.docx` siblings via pandoc alongside the existing `.html` + `.pdf` flow. 44/44 candidates produced valid DOCX (15–18 KB each, `file` reports "Microsoft Word 2007+"). Layer-1 gate `CM-DOCX-EXPORT-SYNC` enforces canonical-doc DOCX sibling presence.

Verification (Mistborn arm64, 2026-05-28)

Gate	Result
`bash scripts/setup.sh --rebuild`	GREEN
`TMUX_BIN=tmux/build-darwin/bin/tmux bash scripts/verify.sh`	**SUMMARY: PASS=45 FAIL=0 SKIP=3**
New tests 45 + 46 + 47 + 48	PASS 6 + 6 + 8 + 9 = 29 / 0 FAIL / 0 SKIP
`bash scripts/tests/meta_test_false_positive_proof.sh`	**43 CAUGHT / 2 ESCAPED (pre-existing P5-M20+P5-M21) / 8 SKIPPED**
`go test ./cmd/workable-items/... -count=3`	**30 PASS / 0 FAIL**
Layer-1 `verify.sh` v1.0.15 gates	All GREEN: 6 conf + 12 propagation + 3 DB + 8 DOCX siblings

The 3 honest SKIPs in the full suite are §11.4.3 topology dispatches:

- `08_oom_score_adj.sh` — Linux-only; SKIPs on Darwin per §11.4.81.
- `12_memory_pressure_under_cap.sh` — destructive; gated by `TMX_TEST_DESTRUCTIVE=1`.
- `32_ssh_dispatch_remote_nezha.sh` — requires nezha host; runs when nezha reachable.

The 2 mutation ESCAPES (P5-M20 + P5-M21) are pre-existing layer-4 test-DESIGN gaps documented in `Issues.md` B3 since v1.0.9 — NOT v1.0.15 regressions.

Verification (nezha Linux ALT 6.12 x86_64, 2026-05-28)

To be populated after PWU-E nezha deploy step.

Honest gaps (out of scope, tracked per §11.4.6)

- **B3 P5-M20 + P5-M21 paired-mutation ESCAPES** continue (pre- existing v1.0.9 layer-4 design gaps in shell-session-resume PWUs). Underlying features GREEN. Closure conditions documented in Issues.md B3.
- **Workable-items live-corpus round-trip** not byte-identical for free-form bodies; golden testdata is. Per §11.4.93 phase-6 migration plan.
- **Workable-items integration into `commit_all.sh` + `sync_issues_docs.sh`** deferred to a follow-up cycle.
- **Upstream PR for Phase 3+ logic** to HelixDevelopment/ HelixConstitution deferred (operator-blocked per `feedback_no_modify_constitution`; project-local implementation satisfies §11.4.93/95 mandate this cycle).
- **TMUX-CH-45 / 46 / 47 / 48 HelixQA bank entries** authored; full bank integration follows the Challenge surface roll-out.
- **`CME-WORKABLE-ITEMS-001`** end-to-end Challenge deferred.

Forensic anchor — RED→GREEN evidence

For tests 46 + 48 specifically (the true RED-first per §11.4.43):

RED state (stock v1.0.14, before fix):

```
$ bash scripts/tests/46_paste_in_physical.sh
  ✗ template MISSING: @clip-read user option
  ✗ template MISSING: prefix+P bind to paste
FAIL: T1: tmux.conf.template is missing the paste-IN bindings
FAIL: T2.1: live server has no @clip-read user-option
FAIL: T2.2: live prefix+P binding is not a paste binding
FAIL: T3: @clip-read not set on live server
FAIL: T4: prefix+P bind does not reference @clip-read
  Tests: PASS=1  FAIL=5  SKIP=0

$ bash scripts/tests/48_modifier_drag_override.sh
  ✗ template MISSING: M-MouseDrag1Pane override
  ✗ template MISSING: S-MouseDrag1Pane override
  ✗ template MISSING: M-MouseDragEnd1Pane pipe
  ✗ template MISSING: S-MouseDragEnd1Pane pipe
FAIL: T1, T2.1, T2.2, T2.3, T2.4, T4
  Tests: PASS=3  FAIL=6  SKIP=0
```

GREEN state (post-fix on same Mistborn this run):

```
$ bash scripts/tests/46_paste_in_physical.sh
PASS: T3: PHYSICAL paste-IN proven — clipboard marker reached
       the tmux pane via @clip-read + paste-buffer
  Tests: PASS=6  FAIL=0  SKIP=0

$ bash scripts/tests/48_modifier_drag_override.sh
PASS: T5: SYSTEM CLIPBOARD (pbcopy/pbpaste) carries all 6 markers
       from the keyboard-equivalent modifier-drag flow
  Tests: PASS=9  FAIL=0  SKIP=0
```

This is the §11.4.43 RED→GREEN cycle proven in a single Mistborn session: tests authored before fix, fail on stock code, pass after fix, captured-evidence per §11.4.5.

Files modified

- `scripts/tmux.conf.template` — A37 conf additions
- `scripts/verify.sh` — Layer-1 gates extended (PWU-A/B/C/D)
- `scripts/tests/meta_test_false_positive_proof.sh` — M46 + M48
- `scripts/challenges/tmux.yaml` — TMUX-CH-45/46/47/48
- `scripts/sync_all_markdown_exports.sh` — NEW (PWU-D)
- `Constitution.md` / `CLAUDE.md` / `AGENTS.md` / `QWEN.md` — §11.4.87..98 short-form blocks (PWU-B)
- `.gitignore` — workable-items DB carve-out (PWU-C)
- `Fixed.md` — A37/A38/A39/A40 closures
- `CHANGELOG.md` — v1.0.15 entry
- `CONTINUATION.md` — §3.16 entry + timestamp bump
- `VERSION` — 1.0.14 → 1.0.15 / versionCode 15 → 16

Files created

- `scripts/tests/45_multiline_copy_physical.sh`
- `scripts/tests/46_paste_in_physical.sh`
- `scripts/tests/47_alt_screen_scroll.sh`
- `scripts/tests/48_modifier_drag_override.sh`
- `scripts/tests/helpers/synthetic_alt_screen_app.py`
- `cmd/workable-items/{main.go,db.go,model.go,parser.go,sync_md_to_db.go,sync_db_to_md.go,diff.go,validate.go,add.go,close.go,schema.sql,db_test.go,roundtrip_test.go}`
- `cmd/workable-items/testdata/{golden_issues.md,golden_fixed.md}`
- `go.mod` + `go.sum`
- `docs/workable_items.db` — SQLite DB, tracked in git per §11.4.95
- `docs/workable-items/{Status.md,Status_Summary.md}` — per §11.4.45/56
- `docs/changelogs/v1.0.15.md` — this file

Submodule pointer bumps

- `constitution/`: `84c948d` → `6828ff2` (19 commits, fast-forward)
- `Containers/`: `fbef9d6` → `2e9ca0e` (17 commits, fast-forward)

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§11.4.92 5-pass change-evaluation summary

1. **Main task verification:** RED→GREEN evidence on tests 46 + 48 captured this run; multi-line keyboard copy proven via pbpaste physical readback.

2. **Regression blast radius:** all 41 existing tests + 4 new tests = 45 PASS, no regression. Meta-test 43 caught, 2 pre-existing escapes only. Existing tests 17 + 44 still PASS (the new bindings don't shadow them).
3. **Cross-feature interaction:** the M-/S-MouseDrag1Pane bindings don't conflict with existing WheelUp / WheelDown / `v` / `y` bindings. @clip-read is a NEW user-option not interfering with @clip. prefix+P doesn't collide with default tmux p (previous- window) because we used uppercase P.
4. **Deep research:** Claude Code's mouse-tracking via CSI ?1003h verified via tmux source; modernc.org/sqlite cross-platform compile verified by build-on-Darwin success.
5. **Anti-bluff confirmation:** every test PASS reads back live server state OR OS clipboard OR Go test framework output. No metadata-only PASS. The RED-first test authoring caught one FAIL-bluff in the test SCRIPT itself (`-N 5 cursor-down` syntax) per §11.4.1 and fixed at source before claiming GREEN.